

| Question | Answer | Mark | Mark scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Additional guidance |
|----------|--------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 18       | Proof  | B1   | for $\overrightarrow{EH} = -\mathbf{b} + \mathbf{a}$ or $\overrightarrow{HE} = -\mathbf{a} + \mathbf{b}$<br>or $\overrightarrow{HG} = \mathbf{a} + \mathbf{c}$ or $\overrightarrow{GH} = -\mathbf{c} - \mathbf{a}$                                                                                                                                                                                                                                                                                                 |                     |
|          |        | B1   | for $\overrightarrow{BC} = -2\mathbf{b} + 2\mathbf{a} + 2\mathbf{c}$<br>or $\overrightarrow{CB} = 2\mathbf{b} - 2\mathbf{a} - 2\mathbf{c}$                                                                                                                                                                                                                                                                                                                                                                         |                     |
|          |        | B1   | for $\overrightarrow{EF} = \mathbf{b} + \frac{1}{2}(-2\mathbf{b} + 2\mathbf{a} + 2\mathbf{c}) (= \mathbf{a} + \mathbf{c})$<br>or $\overrightarrow{FE} = \frac{1}{2}(2\mathbf{b} - 2\mathbf{a} - 2\mathbf{c}) - \mathbf{b} (= -\mathbf{a} - \mathbf{c})$<br>or $\overrightarrow{FG} = \frac{1}{2}(-2\mathbf{b} + 2\mathbf{a} + 2\mathbf{c}) - \mathbf{c} (= -\mathbf{b} + \mathbf{a})$<br>or $\overrightarrow{GF} = \mathbf{c} + \frac{1}{2}(2\mathbf{b} - 2\mathbf{a} - 2\mathbf{c}) (= -\mathbf{a} + \mathbf{b})$ |                     |
|          |        | A1   | for completing proof, $\overrightarrow{EH} = \overrightarrow{FG}$ or $\overrightarrow{HG} = \overrightarrow{EF}$ (so $EFGH$ is a parallelogram)                                                                                                                                                                                                                                                                                                                                                                    |                     |